

Acute Appendicitis: High-Yield Review

Definition and Pathophysiology

Acute appendicitis is the inflammation of the vermiform appendix. The primary initiating event is the obstruction of the appendiceal lumen. In adults, this is most commonly caused by a fecalith, whereas in children, it is typically due to lymphoid hyperplasia following a viral infection.

Clinical Presentation

The classic presentation begins with dull, poorly localized periumbilical pain that migrates to the Right Lower Quadrant (RLQ) over 12 to 24 hours, becoming sharp and well-localized. Associated symptoms include anorexia (very common), nausea, vomiting, and a low-grade fever.

Physical Examination Signs

McBurney's Point Tenderness: Maximal tenderness located one-third the distance from the anterior superior iliac spine to the umbilicus.

Rovsing's Sign: Palpation of the Left Lower Quadrant (LLQ) elicits pain in the RLQ.

Psoas Sign: RLQ pain produced by passive extension of the right hip.

Obturator Sign: RLQ pain with internal rotation of the flexed right hip.

Investigations

Laboratory: Complete Blood Count (CBC) typically shows leukocytosis with a left shift.

Imaging: A CT scan of the abdomen with IV contrast is the gold standard for adults. However, an Abdominal Ultrasound is the preferred initial imaging modality for children and pregnant women to avoid radiation exposure.

Management

Initial management includes keeping the patient NPO (nothing by mouth), administering IV fluids, and starting broad-spectrum IV antibiotics. The definitive treatment is an appendectomy, with the laparoscopic approach being the preferred method for most patients.